

6.6.2.

$$\mu \text{ 的 } 0.95 \text{ 置信区间为 } \left[\bar{x} - \frac{\mu_{1-\frac{\alpha}{2}} \sigma}{\sqrt{n}}, \bar{x} + \frac{\mu_{1-\frac{\alpha}{2}} \sigma}{\sqrt{n}} \right]$$

区间长度小于 k ,

$$\text{则 } \frac{2\mu_{1-\frac{\alpha}{2}} \sigma}{\sqrt{n}} \leq k$$

$$\text{即 } n \geq \frac{4\mu_{1-\frac{\alpha}{2}}^2 \sigma^2}{k^2} = \frac{4 \times 1.96^2 \times 15.376^2}{k^2}$$

6.6.3.

$$(1) \text{ 取 } y_i = \ln x_i$$

$$\text{则 } \mu \text{ 的 } 0.95 \text{ 置信区间为: } \left[\bar{y} - \mu_{1-\frac{\alpha}{2}} \sigma / \sqrt{n}, \bar{y} + \mu_{1-\frac{\alpha}{2}} \sigma / \sqrt{n} \right]$$

$$\text{即 } [-0.98, 0.98]$$

$$(2) E X = e^{\mu + \frac{1}{2}}$$

$$\therefore EX \text{ 的 } 0.95 \text{ 置信区间为 } \left[e^{-0.98 + \frac{1}{2}}, e^{0.98 + \frac{1}{2}} \right] = [0.62, 4.39]$$

6.6.9

(2) 当 $\sigma_1^2 = \sigma_2^2$ 时, μ_1, μ_2 的 0.95 置信区间为

$$\left[\bar{x} - \bar{y} - \sqrt{\frac{n_1+n_2}{n_1 n_2}} s_w t_{1-\frac{\alpha}{2}}(n_1+n_2-2), \bar{x} - \bar{y} + \sqrt{\frac{n_1+n_2}{n_1 n_2}} s_w t_{1-\frac{\alpha}{2}}(n_1+n_2-2) \right]$$

$$\text{即 } [-0.2063, 12.2063]$$

(4) $\frac{b_1^2}{b_2^2}$ 的 0.95 置信区间为

$$\left[\frac{s_x^2}{s_y^2} \frac{1}{F_{1-\frac{\alpha}{2}}(n_1-1, n_2-1)}, \frac{s_x^2}{s_y^2} \frac{1}{F_{\frac{\alpha}{2}}(n_1-1, n_2-1)} \right] = [0.3359, 4.0973]$$

(1) $\sum_{i=1}^n X_i \sim \text{Ga}(n, 1)$

$\therefore \sum_{i=1}^n X_i \sim \chi^2(2n)$

$\therefore P(X_{\frac{\alpha}{2}}^2(2n) \leq \sum_{i=1}^n X_i \leq X_{1-\frac{\alpha}{2}}^2(2n)) = 1-\alpha$

$\therefore$ 人的 1- α 置信区间为 $\left[\frac{X_{\frac{\alpha}{2}}^2(2n)}{2 \sum_{i=1}^n X_i}, \frac{X_{1-\frac{\alpha}{2}}^2(2n)}{2 \sum_{i=1}^n X_i} \right]$

19. (1) 证明: 取 $y = x - \theta$, $\therefore y_i \sim \text{Exp}(1)$

$\therefore f_{X_{(n)}}(y_{(n)}) = n e^{-ny_{(n)}}, y_{(n)} > 0$ 与 θ 无关

(2) $P(c \leq X_{(n)} - \theta \leq d) = \int_c^d n e^{-ny} dy = 1-\alpha$

$\frac{d}{n} \ll d-c$ 取小, 则 $c=0$, $d = -\frac{\ln \alpha}{n}$

$\therefore \theta$ 的 1- α 置信区间为 $[X_{(n)} + \frac{\ln \alpha}{n}, X_{(n)}]$

7.1.1
(1), ~~0.8~~

$$\alpha = P(\bar{X} > 2.6 | H_0) = P\left(\frac{\bar{X}-2}{\sqrt{1/20}} \geq \frac{2.6-2}{\sqrt{1/20}}\right) = 0.0037$$

$$\beta = P(\bar{X} < 2.6 | H_1) = P\left(\frac{\bar{X}-3}{\sqrt{1/20}} < \frac{2.6-3}{\sqrt{1/20}}\right) = 0.0367$$

$$(2) \quad \beta = P\left(\frac{\bar{X}-3}{\sqrt{\frac{1}{n}}} \leq \frac{2.6-3}{\sqrt{\frac{1}{n}}}\right) = 0.01, \quad \text{解得: } n \geq 34$$

$$(3) \quad \alpha = P\left(\frac{\bar{X}-2}{\sqrt{\frac{1}{n}}} \geq \frac{2.6-2}{\sqrt{\frac{1}{n}}}\right) = 1 - \Phi(0.6\sqrt{n})$$

$$\beta = 1 - \Phi(0.4\sqrt{n})$$

当 $n \rightarrow \infty$ 时, α 与 β 都 $\rightarrow 0$

7.1.3. ~~0.8~~ $P(|\bar{X}-6| < c | H_0) = 0.05 \Rightarrow \Phi(2c) = 0.975$
 $c = 0.98$

$$\mu = 6.5 \text{ 时, } \beta = P(|\bar{X}-6| < 0.98 | \mu = 6.5)$$

$$= P\left(-2.96 < \frac{\bar{X}-6.5}{0.5} < 0.96\right) = 0.87$$

7.1.4 $f_{X(n)}(x) = \frac{n x^{n-1}}{\theta^n} \mathbb{I}_{\{0 < x < \theta\}}$

2 $\alpha(\theta) = P(X_{(n)} \leq 2.5 | H_0) = \left(\frac{2.5}{\theta}\right)^n$

$\alpha = \sup \alpha(\theta) = \alpha(3) = \left(\frac{5}{6}\right)^n$

若 $\alpha \leq 0.05$, 则: $n \geq 16.43$, 即 $n \geq 17$

7.2.3, $H_0: \mu = 15$ $H_1: \mu \neq 15$

拒绝域为 $\{u \geq u_{1-\frac{\alpha}{2}}\}$.

而 $u = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = -4.90$ ~~4.90~~ $4.90 > u_{1-\frac{\alpha}{2}}$

∴ 不能认为产品的平均重量仍为 15g

7.2.5 拒绝域为 $\{u \leq u_\alpha\}$ $u_\alpha = -1.65$

∴ $\beta = P(u > -1.65 | \mu \leq 13) = 1 - \Phi\left(-1.65 + \frac{15-\mu}{\sqrt{2.5/n}}\right) \leq 0.05$

∴ $\sup \beta = 1 - \Phi\left(-1.65 + \frac{15-13}{\sqrt{2.5/n}}\right) \leq 0.05$ $n \geq 7$

7.2.13, $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$

拒绝域为 $\{|t| \geq t_{1-\frac{\alpha}{2}}(m+n-4)\}$

而 $t = \frac{\bar{x} - \bar{y}}{\sqrt{s_w(\frac{1}{m} + \frac{1}{n})}} = -0.2056 < t_{0.975}(15)$ ∴ 可以看作一样

7.2.20.

$$H_0: \sigma_1^2 = \sigma_2^2 \quad H_1: \sigma_1^2 < \sigma_2^2$$

拒绝域为 $F \leq F_{\alpha}(m-1, n-1)$

$$\bar{F} = \frac{s_1^2}{s_2^2} = 2.572 > F_{0.05}(9, 9) = 5.9$$

$\therefore$ 不能认为方差相等。